



NUST CHIP DESIGN CENTRE

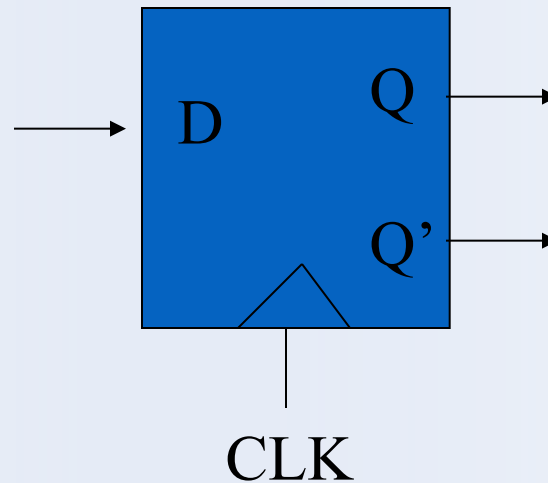
Digital Logic Design

Timing in Sequential Circuits

Sequential Networks

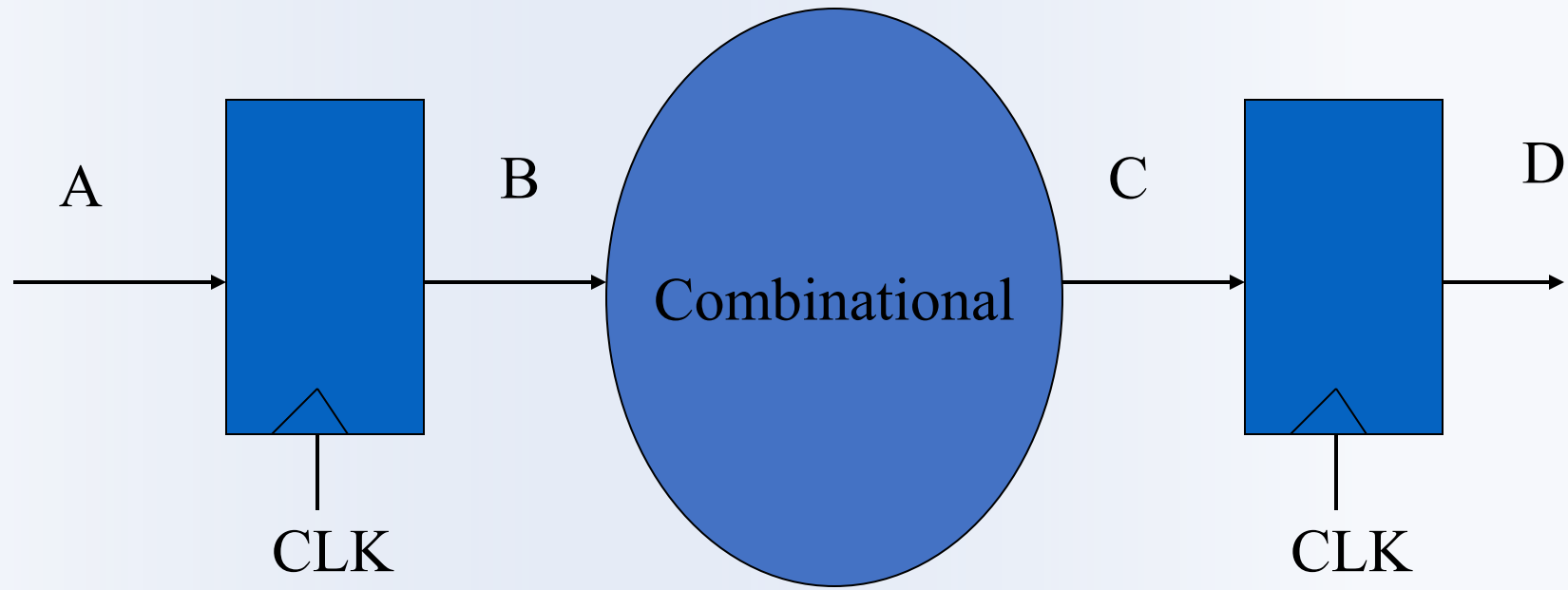


Timing: Setup Time and Hold Time Constraints

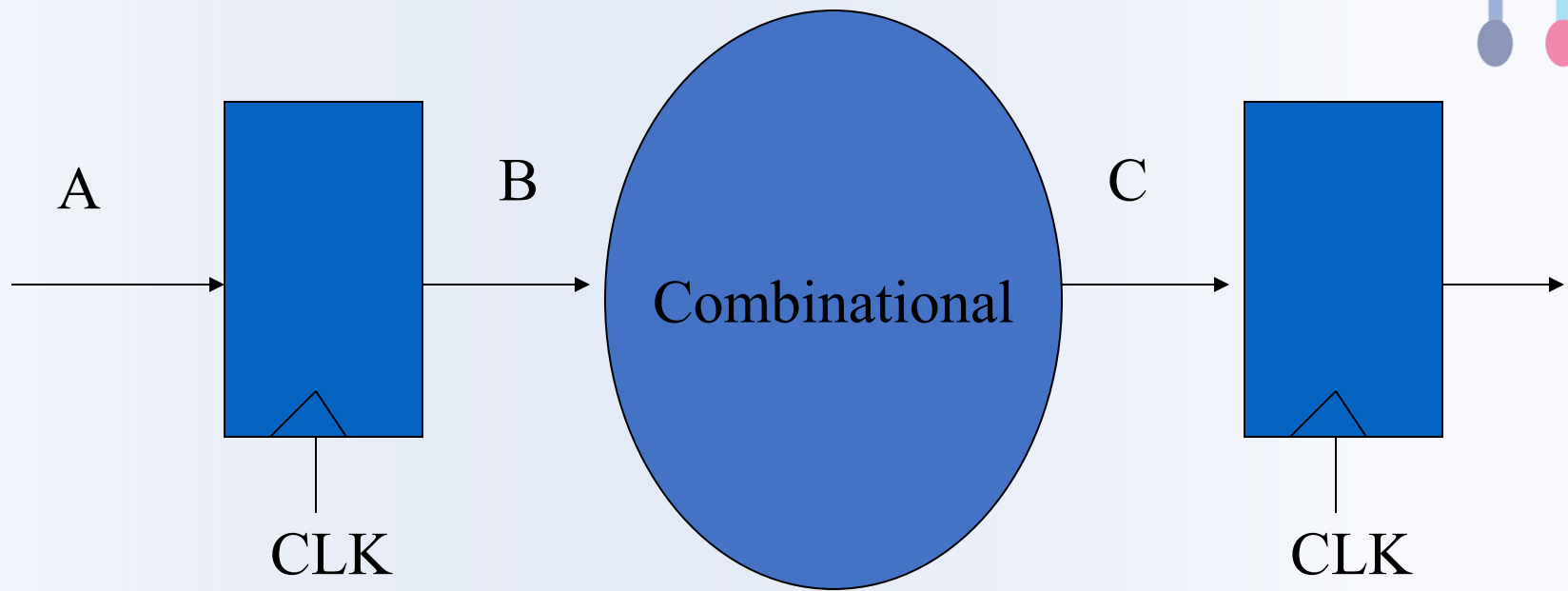




Sequential Networks



A typical sequential network has both a combinational circuit and flip-flops.



$$t_{cq} + t_{comb} + t_{setup} \leq T$$

← Clock period

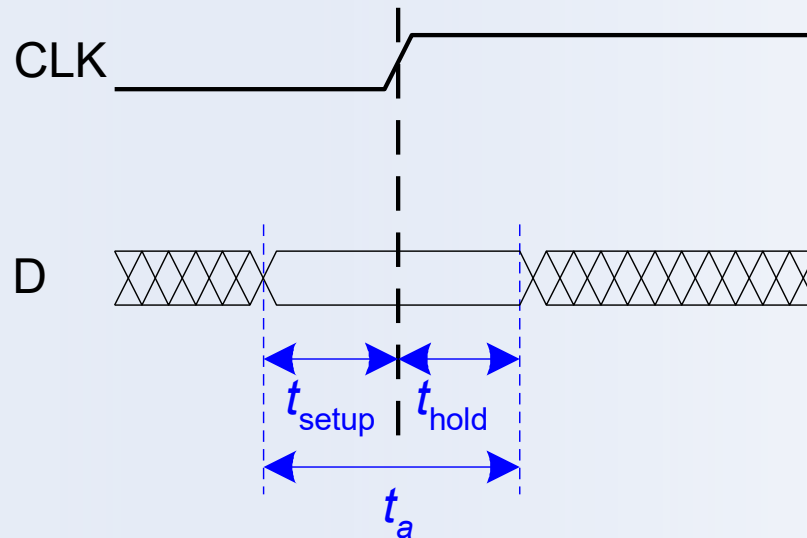
$$t_{hold} < t_{cq} + t_{comb}$$

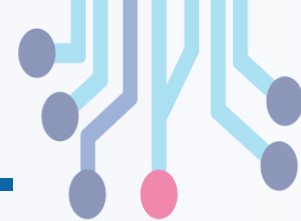
← Shortest path

Input Timing Constraints



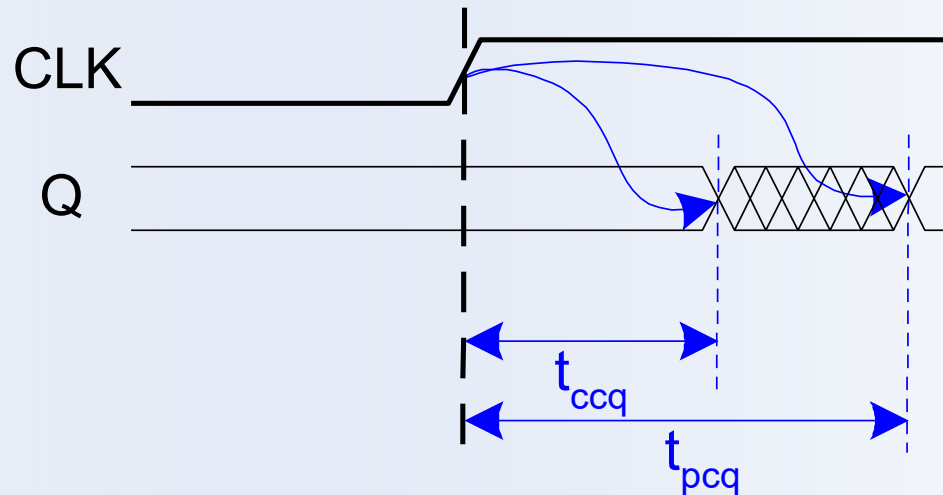
- Setup time: t_{setup} = time *before* the clock edge that data must be stable (i.e. not changing)
- Hold time: t_{hold} = time *after* the clock edge that data must be stable
- Aperture time: t_a = time around clock edge that data must be stable ($t_a = t_{\text{setup}} + t_{\text{hold}}$)

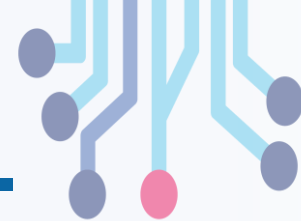




Output Timing Constraints

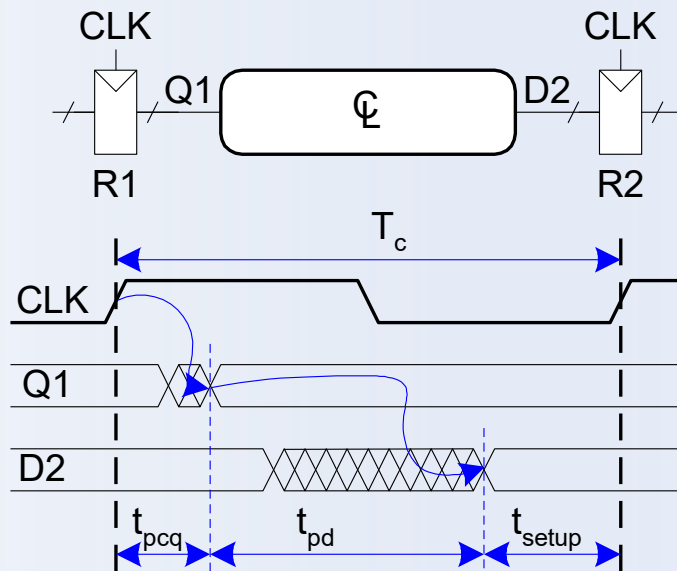
- Propagation delay: t_{pcq} = time after clock edge that the output Q is guaranteed to be stable (i.e., to stop changing)
- Contamination delay: t_{ccq} = time after clock edge that Q might be unstable (i.e., start changing)





Setup Time Constraint

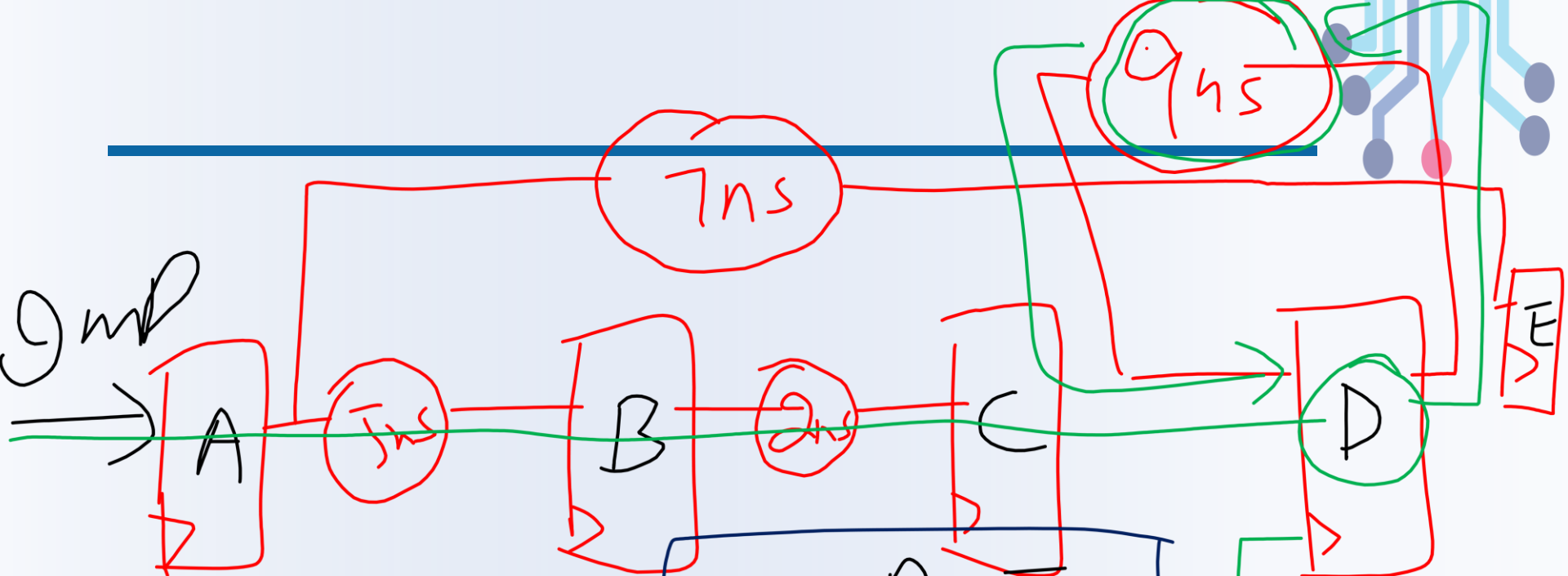
- The setup time constraint depends on the **maximum** delay from register R1 through the combinational logic.
- The input to register R2 must be stable at least t_{setup} before the clock edge.



$$T_c \geq t_{pcq} + t_{pd} + t_{\text{setup}}$$

$$t_{pd} \leq T_c - (t_{pcq} + t_{\text{setup}})$$

t_{pd} : propagation delay

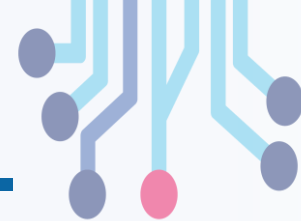


$t_{sup} = 2ns$

$t_{req} = 3ns$

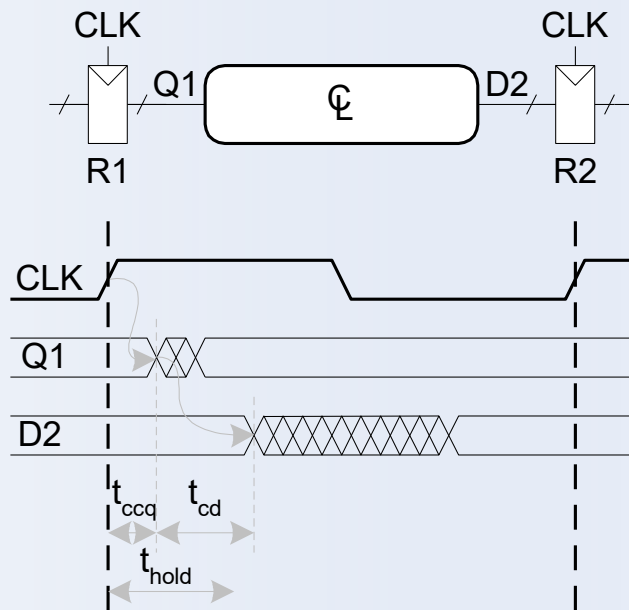
$t_{hold} = 3ns$

- = A, E
- $\Rightarrow$ A, B, C, D
- $\Rightarrow$ C, D
- $\Rightarrow$ D, C, B, A, E



Hold Time Constraint

- The hold time constraint depends on the **minimum** delay from register R1 through the combinational logic.
- The input to register R2 must be stable for at least t_{hold} after the clock edge.



$$t_{\text{hold}} < t_{\text{ccq}} + t_{\text{cd}}$$

$$t_{\text{cd}} > t_{\text{hold}} - t_{\text{ccq}}$$

Timing Analysis



Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

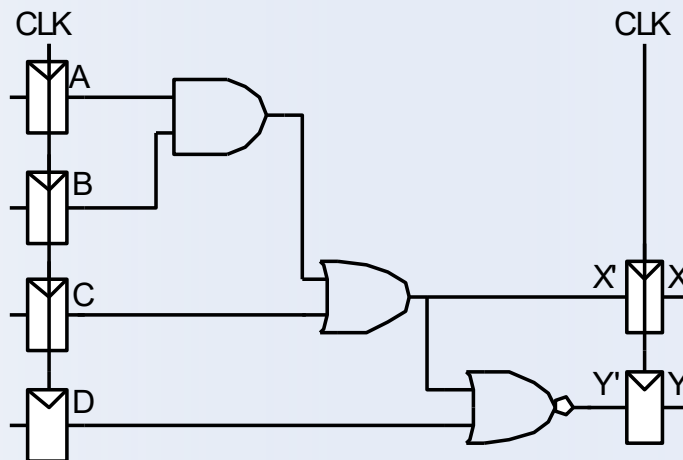
$$t_{pcq} = 50 \text{ ps}$$

$$t_{\text{setup}} = 60 \text{ ps}$$

$$t_{\text{hold}} = 70 \text{ ps}$$

$$t_{pd} = 35 \text{ ps}$$

$$t_{cd} = 25 \text{ ps}$$



$$t_{pd} =$$

$$t_{cd} =$$

Setup time constraint:

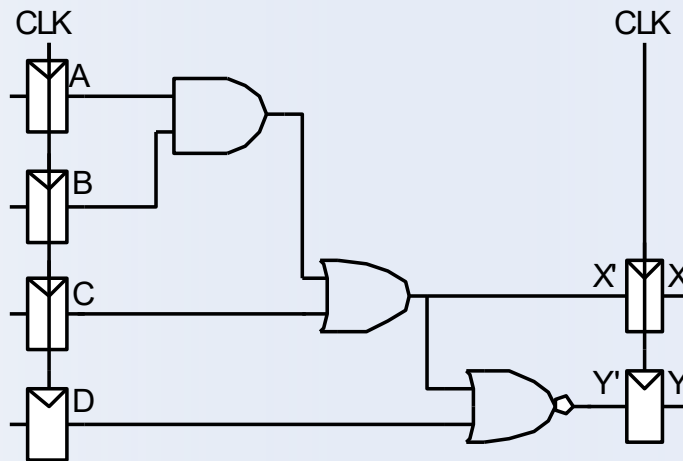
$$T_c \geq$$

$$f_c = 1/T_c =$$

Hold time constraint:

$$t_{ccq} + t_{pd} > t_{\text{hold}} ?$$

Timing Analysis



$$t_{pd} = 3 \times 35 \text{ ps} = 105 \text{ ps}$$

$$t_{cd} = 25 \text{ ps}$$

Setup time constraint:

$$T_c \geq (50 + 105 + 60) \text{ ps} = 215 \text{ ps}$$

$$f_c = 1/T_c = 4.65 \text{ GHz}$$

Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

$$t_{pcq} = 50 \text{ ps}$$

$$t_{\text{setup}} = 60 \text{ ps}$$

$$t_{\text{hold}} = 70 \text{ ps}$$

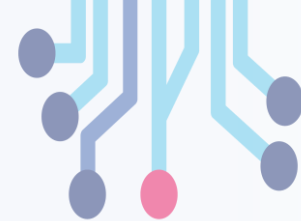
$$t_{pd} = 35 \text{ ps}$$

$$t_{cd} = 25 \text{ ps}$$

Hold time constraint:

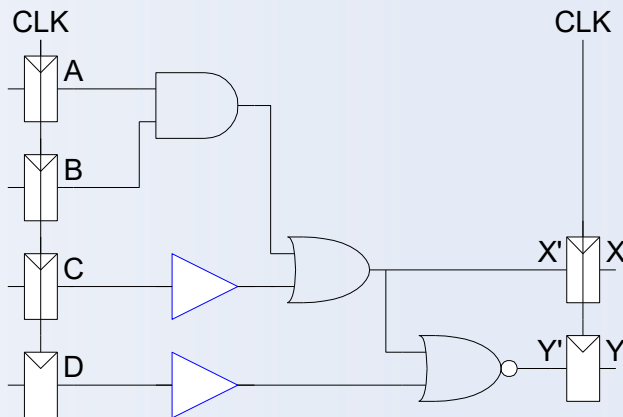
$$t_{ccq} + t_{pd} > t_{\text{hold}} ?$$

$$(30 + 25) \text{ ps} > 70 \text{ ps} ? \text{ No!}$$



Fixing Hold Time Violation

Add buffers to the short paths:



$t_{pd} =$

$t_{cd} =$

Setup time constraint:

$$T_c \geq$$

$$f_c =$$

Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

$$t_{pcq} = 50 \text{ ps}$$

$$t_{\text{setup}} = 60 \text{ ps}$$

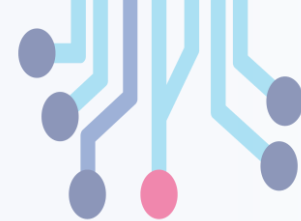
$$t_{\text{hold}} = 70 \text{ ps}$$

$$t_{pd} = 35 \text{ ps}$$

$$t_{cd} = 25 \text{ ps}$$

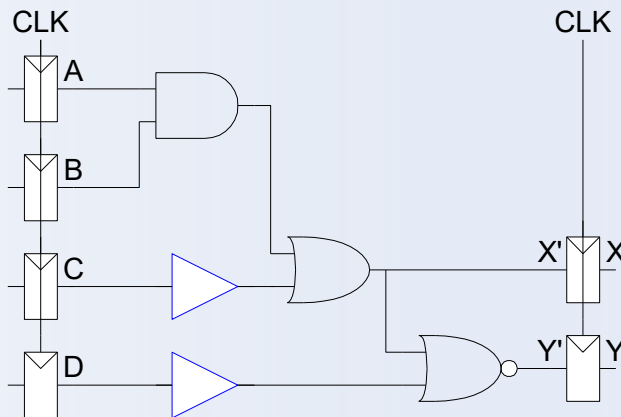
Hold time constraint:

$$t_{ccq} + t_{pd} > t_{\text{hold}} ?$$



Fixing Hold Time Violation

Add buffers to the short paths:



$$t_{pd} = 3 \times 35 \text{ ps} = 105 \text{ ps}$$

$$t_{cd} = 2 \times 25 \text{ ps} = 50 \text{ ps}$$

Setup time constraint:

$$T_c \geq (50 + 105 + 60) \text{ ps} = 215 \text{ ps}$$

$$f_c = 1/T_c = 4.65 \text{ GHz}$$

Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

$$t_{pcq} = 50 \text{ ps}$$

$$t_{\text{setup}} = 60 \text{ ps}$$

$$t_{\text{hold}} = 70 \text{ ps}$$

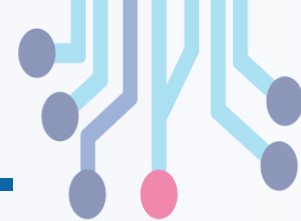
$$t_{pd} = 35 \text{ ps}$$

$$t_{cd} = 25 \text{ ps}$$

Hold time constraint:

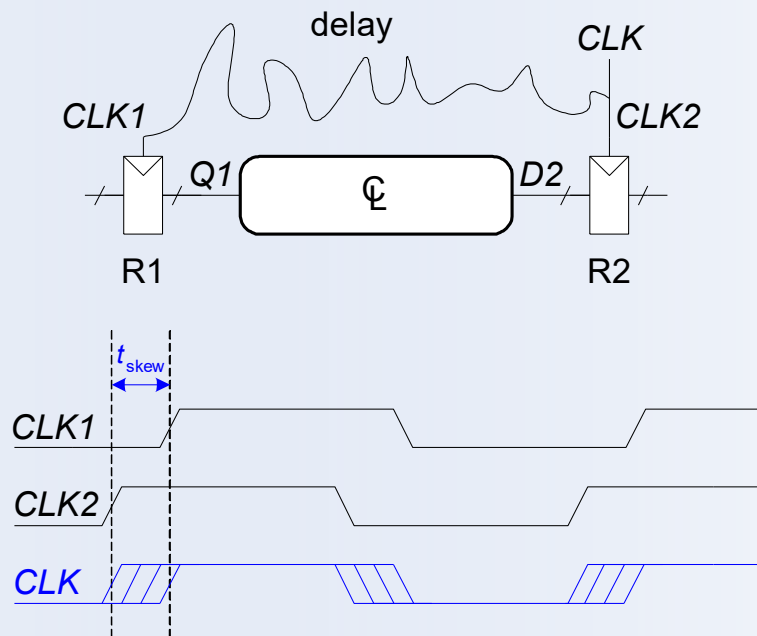
$$t_{ccq} + t_{pd} > t_{\text{hold}} ?$$

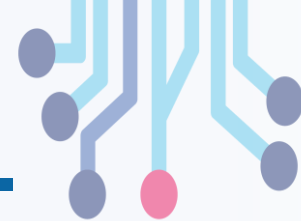
$$(30 + 50) \text{ ps} > 70 \text{ ps} ? \text{ Yes!}$$



Clock Skew

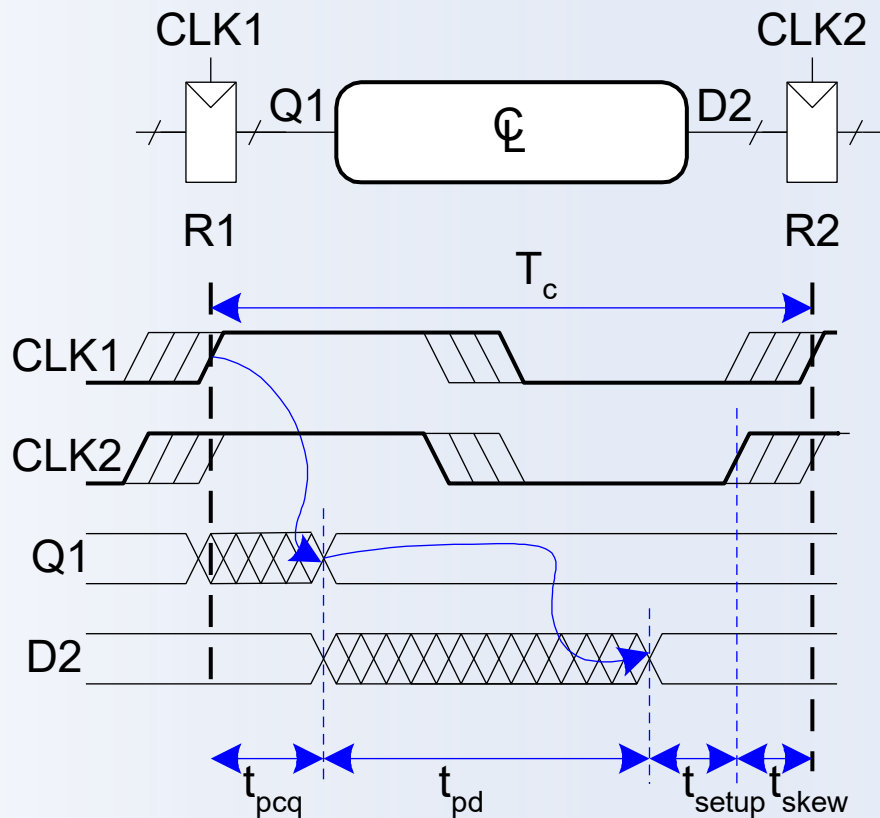
- The clock doesn't arrive at all registers at the same time
- Skew is the difference between two clock edges
- Examine the worst case to guarantee that the dynamic discipline is not violated for any register – many registers in a system!





Setup Time Constraint with Clock Skew

- In the worst case, the CLK2 is earlier than CLK1

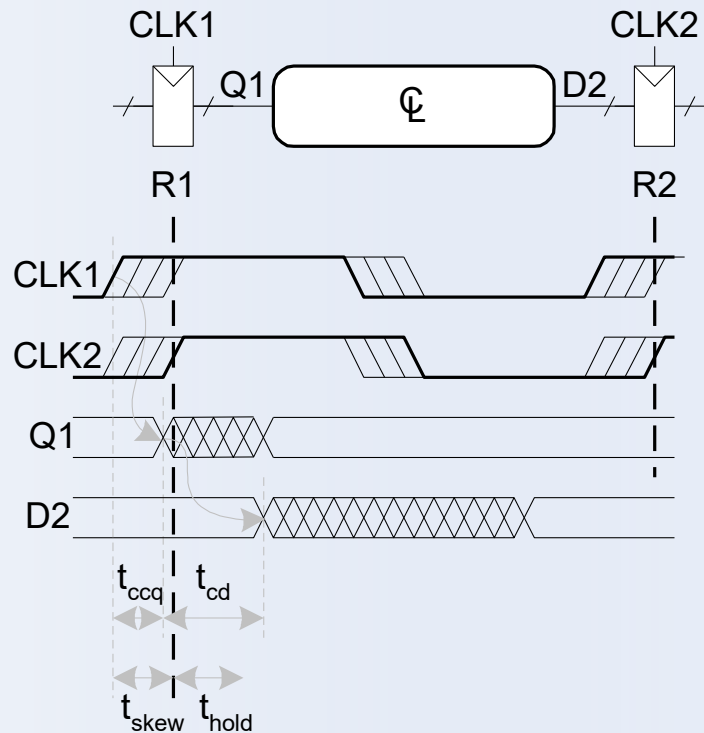


$$T_c \geq t_{pcq} + t_{pd} + t_{setup} + t_{skew}$$
$$t_{pd} \leq T_c - (t_{pcq} + t_{setup} + t_{skew})$$

Hold Time Constraint with Clock Skew



- In the worst case, CLK2 is later than CLK1



$$t_{ccq} + t_{cd} > t_{hold} + t_{skew}$$

$$t_{cd} > t_{hold} + t_{skew} - t_{ccq}$$

Timing and Retiming



- Retiming: Adjust the clock skew so that the clock period can be reduced.
- Add a few more examples on timing and retiming.

Fixing Setup Time Violation



1. Increase clock period
2. Optimize datapath
 - a. Pipeline
 - b. Logic optimization
 - c. Buffer removal
 - d. Routing optimization
3. Use faster cells
4. Relax constraints for non-critical paths

Fixing Hold Time Violations



1. Add delay to datapath
 - a. Insert buffers
 - b. Interconnect delay
2. Reduce clock skew
 - a. Clock tree optimization
3. Increase hold time requirements
 - a. Use of retiming buffers
 - i. Retiming involves repositioning registers in the circuit to balance the timing constraints. By shifting flip-flops, it's possible to meet the required hold time without impacting overall circuit performance.
 - b. Insert hold buffers
4. Adjust operating conditions
 - a. Lowering voltage
 - b. Temperature adjustment
5. Change Flip-flop type